J. Geomag. Geoelectr., **48**, 993–1000, 1996

Palaeomagnetism of the Rajmahal Traps of India: Implication to the Reversal in the Cretaceous Normal Superchron

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(Received July 20, 1995; Revised March 1, 1996; Accepted March 4, 1996)

The Upper Gondwana Rajmahal Traps in northeast India were studied over the last four decades and these are referred to as normally magnetized rocks. Our study of these rocks from thirty-five sites has brought into light a palaeomagnetic record revealing the existence of a reverse magnetization event also. In the light of the $^{40}Ar/^{39}Ar$ age data of the Rajmahal Traps and the geomagnetic polarity time scale during the middle Cretaceous period, the observed magnetic reversal in the Rajmahal Traps has been interpreted to constitute a geomagnetic reversal in the Cretaceous Normal quiet magnetic interval (Cretaceous Normal Superchron, K-N). Similar short magnetic reversals have also been reported at other periods in this superchron from some other places.

1. Introduction

The Rajmahal Traps are spread over an area of approximately 4000 sq.km located in the states of Bihar and West Bengal in NE India. These traps and the associated dykes and sills together with the Sylhet and Bengal Traps represent the only known volcanic suite of rocks in the Upper Gondwana succession of late Mesozoic age (Pascoe, 1959). Magmatic activity in eastern India is appreciably extensive (Baksi *et al.*, 1987) and Mahoney *et al.* (1983) have suggested that the Kerguelen hotspot appears to have been the source for the Rajmahal Traps, as the Indian subcontient moved across the hotspot subsequent to break up of the Gondwanaland. Alternatively it has also been suggested that the Crozet hotspot is responsible for the Rajmahal volcanism (Curray and Munasinghe, 1991). This flood basalt volcanism in eastern India is related to the Aptian extinction event by Rampino and Stothers (1988).

The Rajmahal Traps have a maximum thickness of 600 m, comprising at least 15 distinct andesite flows which are 20–75 m in thickness including one flow of pitchstone. Subsurface extension of the basalt has been recorded below the Gangetic alluvium in the north as well as to the east and south below the Bengal basin, indicating that a greater part (nearly 2/3) of these volcanics are buried in the depressed eastern Indian shield, where several N-S basement faults have been mapped by seismic and other geophysical methods.

On the basis of palaeontological evidence the Gondwana System has been divided into the lower Gondwana Sequence (Glossopteris Flora) and the upper Gondwana Sequence (Ptillophyllum Flora) which hosts the Rajmahal Traps (Wadia, 1953; Krishnan, 1960; Pascoe, 1959; Tiwari and Tripathi, 1995). K-Ar ages of these traps (McDougall and McElhinny, 1970; Agarwal and Rama, 1976) range from 75–110 Ma. Baksi *et al.* (1987) have proposed that flood basalt volcanism in eastern India was of limited duration probably spanning as little as 10 Ma at Ca. 115 Ma. $^{40}Ar/^{39}Ar$ age of a single sample suggests a minimum age of 110 Ma for the Rajmahal volcanism (Dalrymple and Lanphere, 1974). Baksi (1995) reported a $^{40}Ar/^{39}Ar$ age of 117 Ma with a duration of 2 Ma. In the Rajmahal Hills no younger sedimentary strata overlie the Rajmahal Group except for the intertrappeans, superficial laterite and alluvium. However, in the Bengal basin to the east, boreholes drilled for petroleum exploration have brought to light a sequence of Cenozoic and Cretaceous rocks overlying these traps.

2. Palaeomagnetism

Oriented samples from thirty-five sites as shown in Fig. 1 from all over the Rajmahal Traps were studied for the palaeomagnetic signature in it. The specimens from all the sites exhibit uniform intensity and susceptibility with very good grouping of the NRM vectors from all the sites which confine to normal and reverse magnetizations. Component structures have been evaluated using laboratory AF and thermal demagnetization techniques. Two to three specimens from each site were pilot AF demagnetized in progressively increasing peak alternating fields upto 100 mT. These studies reveal very good stability of the remanent vector with median destructive fields (mdf) between 10–30 mT without any superposed components of magnetization. Typical response of the specimens to AF demagnetization is shown in Fig. 2(a). A field of 20 mT is found to be sufficient to reveal the characteristic remanent magnetic vector in these specimens.

In a similar fashion two to three specimens from each site were studied on a pilot basis using thermal demagnetization technique. The sepcimens were subjected to increasing temperatures in steps upto 600°C and observed their response to heating. The specimens exhibit low blocking temperatures between 400–450°C after which the vectors make large migration at higher temperature treatments. Typical behaviour

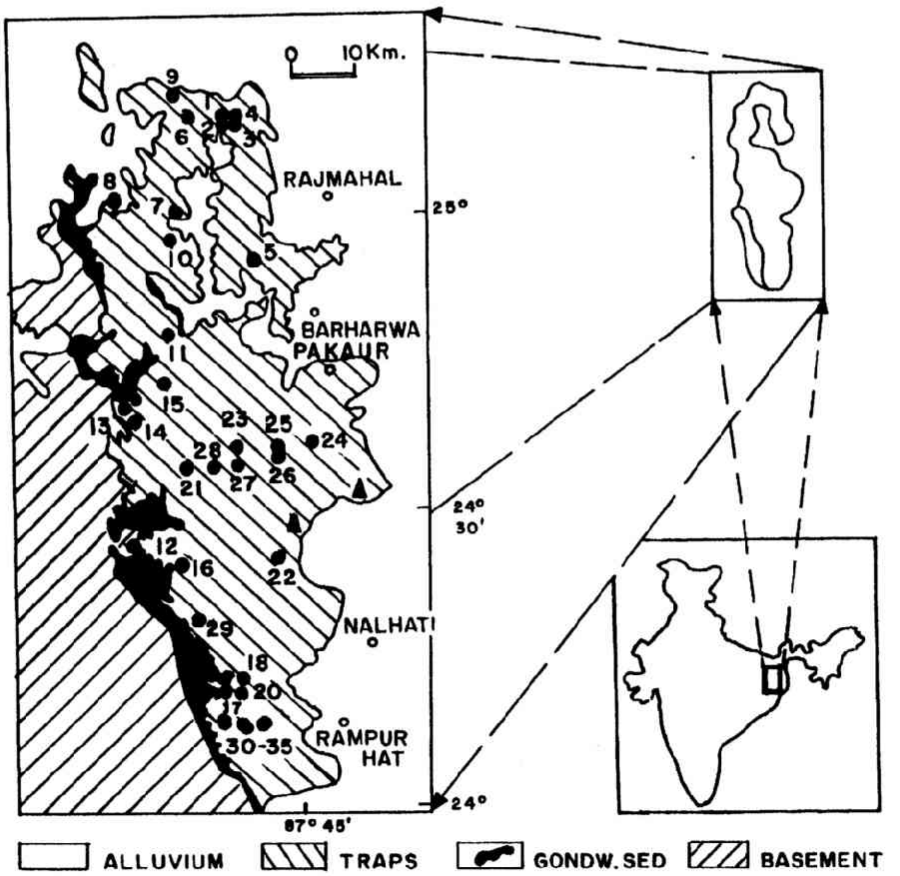


Fig. 1. Geological map of the Rajmahal Traps in northeastern India showing the palaeomagnetic sampling site locations (closed circles). Location of samples for $^{40}Ar/^{39}Ar$ dating are also shown (filled triangles).

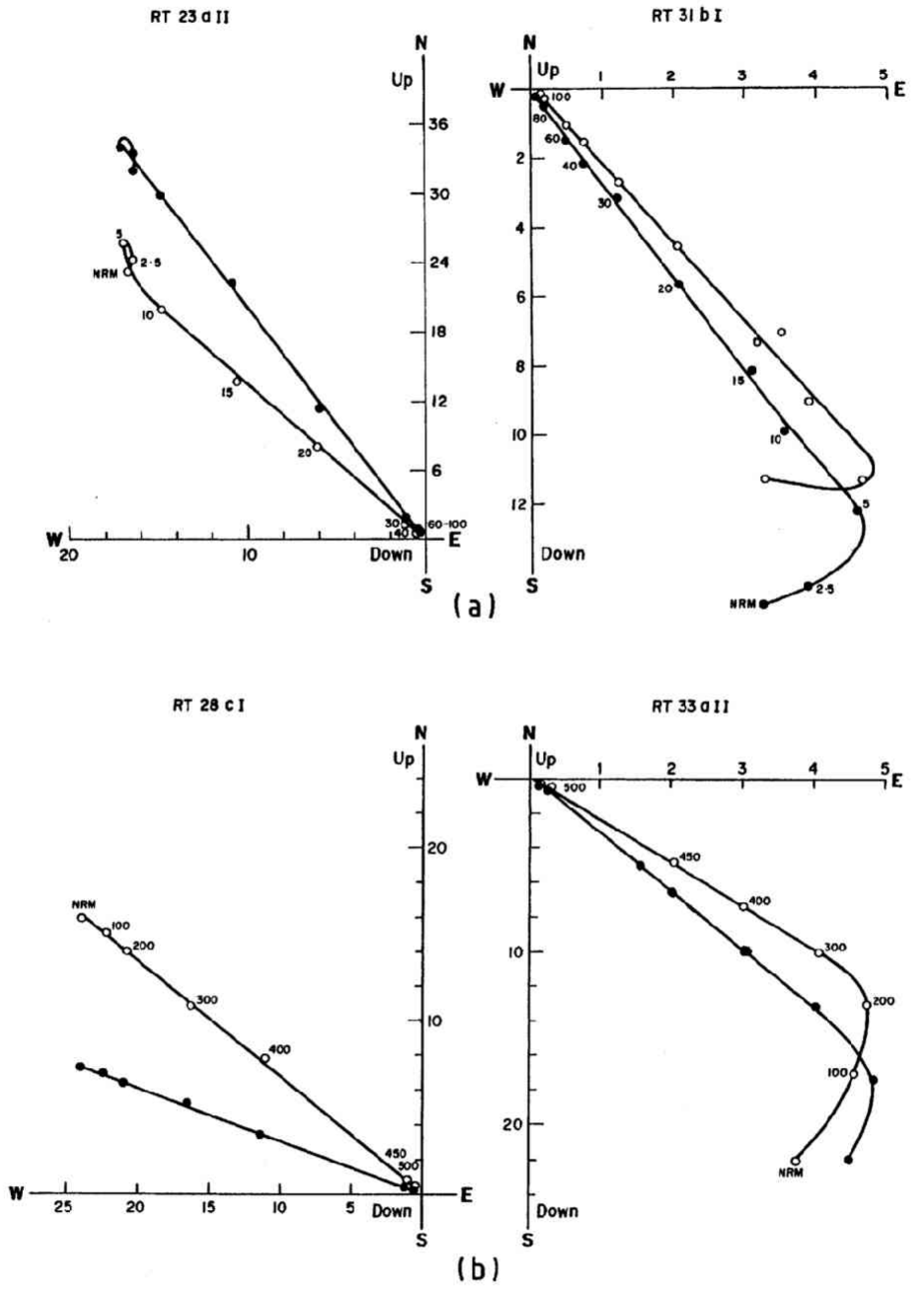


Fig. 2. Behaviour of Rajmahal Traps to laboratory AF (a) and thermal (b) treatment on orthogonal projection. Open circles represent projection of the resultant vector on the E-W horizontal plane and closed circles on the N-S vertical plane. Intensities are in A/m.

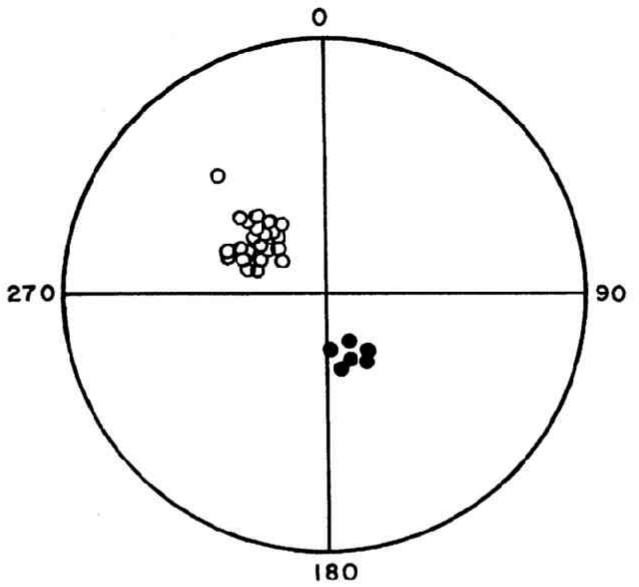


Fig. 3. Stereographic plot showing the site characteristic mean remanent direction vectors of the Rajmahal Traps. Open (closed) circles are the upward (downward) inclinations.

of the specimens to thermal treatment is shown in Fig. 2(b). A temperature of 400°C is found to be sufficient to recover the characteristic component is these specimens. From the above laboratory demagnetizations and other magnetic properties it is inferred that titanium rich titanomagnetite is the magnetic carrier in these rocks as revealed by granulometric studies by Deutsch *et al.* (1980) and Radhakrishnamurty (1990).

As a result of laboratory demagnetization grouping of the remanent magnetic directions of the sites improved largely. Characteristic directions of magnetization were determined using Zijderveld plots, vector plots and normalised intensity decay diagrams. Site mean directions were computed using Fisher (1953) statistical methods and is shown in Fig. 3. Out of 35 sites investigated 29 sites exhibit normal polarity and 6 sites reverse polarity with a mean direction of $D_m = 316.9$, $I_m = -64.1$, $K = 105.77$, $A_{95} = 2.33$, $N = 34$ (after excluding one site with shallow inclination, Fig. 3). Sitewise palaeomagnetic parameters are listed in Table 1.

Palaeomagnetic properties of the Rajmahal Traps have been recorded by several workers over the last four decades (Clegg *et al.*, 1958; Radhakrishnamurty, 1970; McDougall and McElhinny, 1970; Klootwijk, 1971; Poornachandra Rao *et al.*, 1992). Independent studies on these traps have resulted in almost identical remanent magnetic vectors and on rockmagnetic count Radhakrishnamurty (1970, 1990) opined that these rocks possess only one component and hence need no laboratory demagnetization to unfold their characteristic remanent magnetism. A summary of all available results is given in Table 2 and mean direction and VGP for the Rajmahal Traps is evaluated by averaging out all these results.

3. Discussion

Previous workers had reported only normal magnetization except Klootwijk (1971) and Poornachandra Rao *et al.* (1992) who uncovered reverse magnetization also in their studies. From these investigations it is clear that the reversely magnetized flows are located at the southern part of the Rajmahal Trap outcrop. McDougall and McElhinny (1970) also collected samples from this part but failed to identify the reversely magnetized horizon.

Table 1. Site mean remanent magnetic data of Rajmahal Traps, NE India.

S. No.	Lat. (N)	Long. (E)	N	n	D_m	I_m	K	A_{95}	λ_p (°N)	L_p (°W)
1	25.22	87.75	3	18	324	–56	170.9	6.2	19.8	62.5
2	25.22	87.75	3	17	317	–58	89.3	8.6	14.5	59.0
3	25.22	87.75	3	16	317	–59	102.0	8.0	13.5	59.8
4	25.22	87.75	3	18	319	–63	175.4	6.1	11.3	63.7
5	24.96	87.75	6	36	316	–60	105.4	5.6	12.9	59.3
6	25.20	87.67	3	16	302	–63	94.3	8.3	02.9	55.3
7	25.05	87.62	3	18	321	–57	44.2	12.2	17.3	61.4
8	25.05	87.48	3	18	318	–60	57.8	10.6	12.9	61.2
9	25.25	87.65	5	28	299	–58	47.0	9.1	04.2	49.4
10	25.00	87.60	3	15	320	–37*	84.8	10.8	30.1	48.7
11	24.50	87.60	3	18	315	–76	307.7	4.6	–5.0	73.9
12	24.50	87.50	3	16	289	–65	289.9	4.8	–5.7	51.4
13	24.70	87.51	3	17	334	–63	114.3	3.0	17.1	72.9
14	24.70	87.51	3	18	296	–67	20.6	17.8	–3.5	57.2
15	24.75	87.60	3	18	317	–53	44.9	12.1	18.7	56.0
16	24.45	87.61	5	31	301	–67	128.2	5.5	–0.3	58.4
17	24.25	87.50	3	18	297	–62	196.0	5.8	01.0	52.3
18	24.20	87.65	3	17	323	–62	68.0	12.0	13.9	65.6
19	24.20	87.65	4	18	293	–58	14.0	26.4	01.2	46.5
20	24.25	87.65	6	32	329	–64	104.4	6.1	15.3	70.5
21	24.60	87.60	3	18	313	–65	408.2	4.0	07.2	62.5
22	24.45	87.75	3	16	321	–73	110.5	7.7	01.1	73.1
23	24.65	87.65	3	16	310	–63	38.0	13.1	07.9	58.3
24	24.65	87.85	3	17	314	–65	909.1	2.7	08.0	62.5
25	24.65	87.80	3	16	309	–61	13.6	21.9	08.3	56.6
26	24.65	87.80	3	17	318	–63	192.3	5.8	11.0	63.3
27	24.60	87.40	6	31	306	–64	37.2	9.4	04.1	58.1
28	24.60	87.65	4	23	324	–65	92.0	7.3	11.7	68.2
29	24.35	87.60	3	18	322	–60	138.9	8.4	15.9	62.9
30	24.25	87.65	3	15	166	+67	28.9	15.0	15.0	82.9
31	24.20	87.65	3	17	172	+71	103.1	9.8	10.8	87.7
32	24.21	87.65	3	17	139	+67	555.6	3.4	07.4	66.7
33	24.20	87.65	3	16	155	+65	168.1	6.2	11.6	70.4
34	24.20	87.65	3	18	147	+64	227.3	5.4	13.8	69.0
35	24.20	87.60	4	24	153	+71	227.3	4.2	07.1	77.2
	Mean (34 sites)				317	–64	105.8	2.3	09.4	63.4

Lat. = Latitude of site, Long. = Longitude of site, N = No. of samples, n = No. of specimens, D_m = Mean Declination, I_m = Mean Inclination, K = Precision parameter, A_{95} = Radius of confidence, λ_p = Latitude of VGP, L_p = Longitude of VGP.
*Not included in the mean.

Among the different basaltic flows comprising the Rajmahal Traps, no correlation of these flows at different localities is available making it difficult to define precisely the transition of magnetic reversal. Cleaned directions of the reversely polarised rocks record somewhat steeper inclinations in comparison to the normal ones (Klootwijk, 1971; Poornachandra Rao *et al.*, 1992) (see also Fig. 3). Our field observations reveal that the Gondwana sediments (Barakars), over which the Rajmahal Traps rest directly have an easterly dip in the western and southern regions and one can see such situations at several locations. In view of the above field evidence and steeper upward inclination in the reversely magnetized flows, we believe that the transition of the polarity inversion is from reverse to normal.

The $^{40}Ar/^{39}Ar$ date of the Rajmahal Traps at 117 Ma with a duration of 2 Ma (Baksi, 1995) with a magnetic reversal (Klootwijk, 1971; Poornachandra Rao *et al.*, 1992) places it in the Cretaceous Normal

Table 2. Remanent magnetic data of Rajmahal and Sylhet Traps.

M	N	D_m	I_m	K	A_{95}	λ_p (°N)	L_p (°W)	δ_p	δ_m	Reference
					Rajmahal Traps					
3	17	328	–64	—	2	13	70	—	—	Clegg *et al.* (1958)
15	120	323	–64	—	4	12	67	5.0	6.5	Radhakrishnamurty (1970)
8	16	310	–67	187	4	3	62	5.4	6.5	McDougall and McElhinny (1970)
25	158	315	–65	60	4	7	63	4.5	6.0	Klootwijk (1971)
34	120	317	–64	106	2	9	63	3.0	3.7	This study
*Mean 82 sites		316	–65	833	2	8	64	3.0	3.5	
					Sylhet Traps					
—	36	332	–59	—	7	16	60	8.0	10.5	Athavale *et al.* (1963)
3	18	319	–62	62	7	12	59	8.7	11.2	Poornachandra Rao *et al.* (1993)
		235	–62	55	5					

M = Number of sites, N = Number of samples, D_m = Mean Declination, I_m = Mean Inclination, K = Precision parameter, A_{95} = Radius of circle of confidence, λ_p = Latitude of VGP, L_p = Longitude of VGP, δ_p = Major axis of error Ellipse, δ_m = Minor axis of error Ellipse.

*The results by Clegg *et al.* (1958) were not included in the mean for the Rajmahal Traps as they were not AF demagnetized.

Superchron (KN). The polarity scale of magnetic reversals during Mesozoic era, shown in Fig. 3 (Harland *et al.*, 1990) shows frequent reversals during the lower and upper Cretaceous periods. During the middle Cretaceous a very long normal interval prevailed between 124.5 and 83 Ma, except for two short events (M0 and M1) (Harland *et al.*, 1982, 1990; Piper, 1987). However, the GPTS after Lowrie and Ogg (1986) indicate Chron M0 between 113–114 Ma, but caution a 10 Ma uncertainty in absolute ages of polarity chrons. There are several studies in the recent past that reveal short intervals of reversed polarities which constitute reversals in this superchron. Vanden Berg *et al.* (1978) and Tarduno *et al.* (1992) observed reversed magnetizations at 115 and 105 Ma respectively from Umbrian Appenines in Northern Italy. A 96 Ma old gabbro from Smith Island, Canada, has been found to have recorded normal and reversed magnetizations by Archibald and Irving (1990). DSDP cores from Pacific and Atlantic Oceans revealed reversed polarity zones which were well correlated with the ISEA interval of Appenines at 115 Ma (Tarduno, 1990). Similarly the Rajmahal Traps reversal falling in the superchron at 117 ± 1 Ma would constitute one more reversal between 118–116 Ma. Figure 4 summarises all the reversals along with M0 and M1 within this superchron during the middle Cretaceous between 124.5 and 83 Ma which were observed during this period.

Palaeolatitude of the Rajmahal Traps ($\lambda_m = 43°S$) revealed by the present study is also consistent with their age. India broke apart from Antarctica at 120 Ma ago and crossed the Kerguelen hotspot (located in a quasi-fixed mantle reference frame at 50°S) which was believed to have provided the material for the Rajmahal Traps at 117 ± 1 Ma, while India was located at 43°S latitude on its northward journey. Ages of 124 and 83 Ma are inconsistent with observed reversal in the Rajmahal Traps with the pattern of the progressive northern movement of India. Palaeomagnetic studies of the Rajmahal and the Sylhet Traps (Table 2) also support the hypothesis that magmatic activity in eastern India is a part of a widespread episode commenced at 119–118 Ma. Therefore, in view of the above evidences it is suggested that the reversal observed in the Rajmahal Traps constitutes a magnetic reversal between 118–116 Ma in the Cretaceous Normal Superchron (KN) during which period magnetic reversals are also frequent similar to other mixed polarity periods.

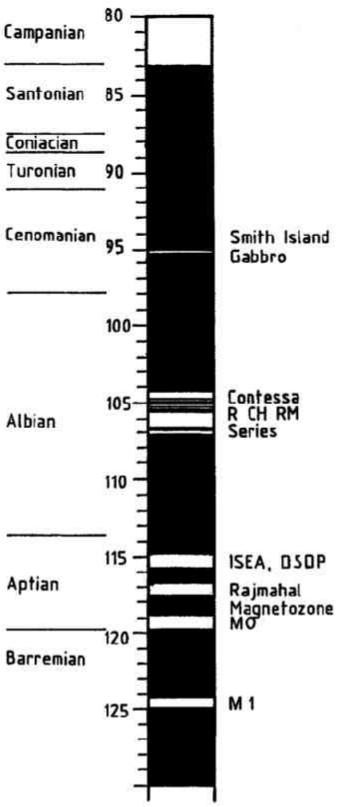


Fig. 4. Geomagnetic Polarity Time Scale (GPTS) showing the geomagnetic reversals during the Cretaceous period (after Harland *et al.*, 1990). Smith Island Gabbro (Archibald and Irving, 1990); Contessa R ChRM Series (Tarduno *et al.*, 1992); ISEA (Vanden Berg *et al.*, 1978) and DSDP Sites (Tarduno, 1990).

We are thankful to the Director, National Geophysical Research Institute, Hyderabad, for his keen interest in these studies and kind permission to publish these results. We are thankful to Dr. M. V. Subba Rao for his support during field work.

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